Joint Radio-Environment Map Construction for THz ISAC via Conditional Flow Matching

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***Abstract*—Terahertz (THz) integrated sensing and communication (ISAC) requires accurate radio-environment awareness despite severe blockage and sparse measurements. This paper investigates joint reconstruction of received signal strength (RSS) maps and obstacle occupancy grids from sparse RSS samples. We formulate the task as conditional generation of a two-channel radio-environment map. A conditional flow matching (FM) framework transports Gaussian noise to the conditional map distribution, guided by sparse RSS observations and sensing masks. The method jointly reconstructs RSS fields and obstacle occupancy through one-step or few-step ordinary differential equation (ODE) sampling. Ray-tracing experiments show lower MSE than DDPM, DDNM, and RadioUNet across the tested sensing ratios. At a sensing ratio of 0.01, RSS and occupancy-grid MSE decrease by 57.8% and 62.4%, respectively, relative to DDPM. These results support FM as an efficient reconstruction component for THz ISAC.**

***Index Terms*—Terahertz ISAC, radio map, conditional flow matching, occupancy inference, blockage awareness.**

I. Introduction

Terahertz (THz) integrated sensing and communication (ISAC) is expected to support ultra-high-rate transmission and high-resolution environmental awareness in future 6G networks [1]. However, THz links are sensitive to path loss, molecular absorption, directional misalignment, and obstacle blockage [2], [3]. Reliable THz communication therefore requires both signal-strength prediction and knowledge of surrounding obstacles. We use the term radio-environment map for this joint representation: the RSS map describes link quality, and the obstacle occupancy grid describes environmental blockage.

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Constructing dense radio-environment maps from sparse measurements is severely ill posed. In practice, only a limited number of sensors, user equipment devices, or probing nodes report local RSS values. Most grid cells remain unobserved, and obstacle occupancy is not directly measured. THz propagation also combines smooth large-scale attenuation with sharp blockage-induced discontinuities, limiting the accuracy of simple interpolation.

Existing methods face limitations in this joint THz ISAC setting. Classical interpolation and Kriging exploit spatial smoothness but do not directly infer obstacle occupancy. Deterministic networks such as RadioUNet learn mappings from sparse observations to dense maps, but may over-smooth uncertain regions and provide only point estimates. Diffusion models represent richer conditional distributions, yet conventional samplers often require hundreds or thousands of steps. This cost motivates more efficient reconstruction for latency-sensitive beam management and network control.

To address these limitations, we propose a conditional flow matching framework for joint radio-environment map reconstruction. The model learns a velocity field that transports Gaussian noise to conditional RSS–occupancy maps, given sparse RSS observations and sensing masks. One-step or few-step ODE sampling exploits the learned relationship between radio propagation and environmental blockage. Experiments on a ray-tracing THz dataset show lower reconstruction MSE than DDPM, DDNM, and RadioUNet across different sensing ratios. The results support its potential as an efficient reconstruction component for THz ISAC.

II. Related Work

Radio map construction has been studied using interpolation, neural reconstruction, and generative modeling.

Kriging exploits spatial smoothness but struggles with blockage-induced discontinuities. RadioUNet improves RSS estimation through deterministic neural prediction [7]. Diffusion models support conditional generation, while flow matching learns an ODE velocity field for efficient sampling [9]. Recent work such as RadioFlow [12] applies flow matching to radio map construction. Here, we focus on joint RSS and obstacle-occupancy reconstruction for THz ISAC using sparse RSS observations.

III. System Model

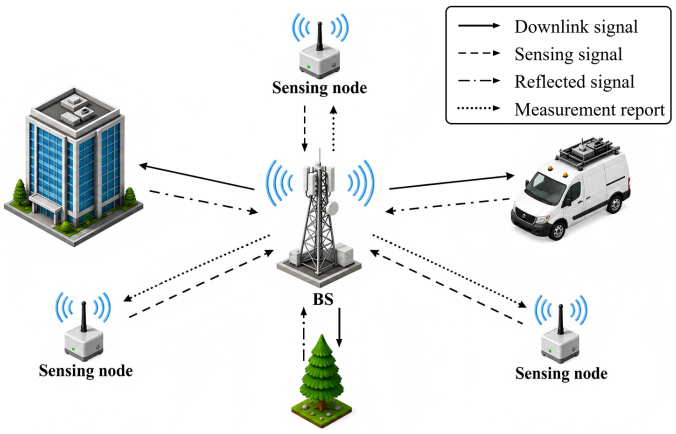


Fig. 1: THz ISAC layout with a central BS, sparse RSS sensors, and blockage-inducing obstacles.

We consider a sensor-assisted THz ISAC system over a two-dimensional area $\mathcal{A} \subset \mathbb{R}^2$, as shown in Fig. 1. A base station (BS) with a directional antenna array transmits THz communication and probing signals. A limited number of sensing nodes or user equipment devices report local RSS measurements through a control channel. Buildings, trees, and vehicles cause blockage, reflection, and scattering. The BS reconstructs a dense radio-environment map from these observations to support blockage-aware beam selection, handover, and resource allocation.

We discretize $\mathcal{A}$ into an $L \times W$ grid. The RSS map $\mathbf{P} \in \mathbb{R}^{L\times W}$ gives the received signal strength at each cell. The occupancy map $\mathbf{S} \in [0,1]^{L\times W}$ represents obstacles, with values near zero and one indicating free and occupied regions, respectively. We define the joint radio-environment map as

$$\mathbf{X} = [\mathbf{P}; \mathbf{S}] \in \mathbb{R}^{2\times L\times W}, \tag{1}$$

where the two channels represent signal strength and obstacle occupancy, respectively. This representation links radio propagation to the environmental structures that shape it.

Only a subset of grid cells is observed. Let Ω denote the sensing locations and $\mathbf{M} \in \{0,1\}^{L\times W}$ the binary sensing mask, with $[\mathbf{M}]_{i,j} = 1$ for $(i,j) \in \Omega$ and zero otherwise. The sparse RSS observation is

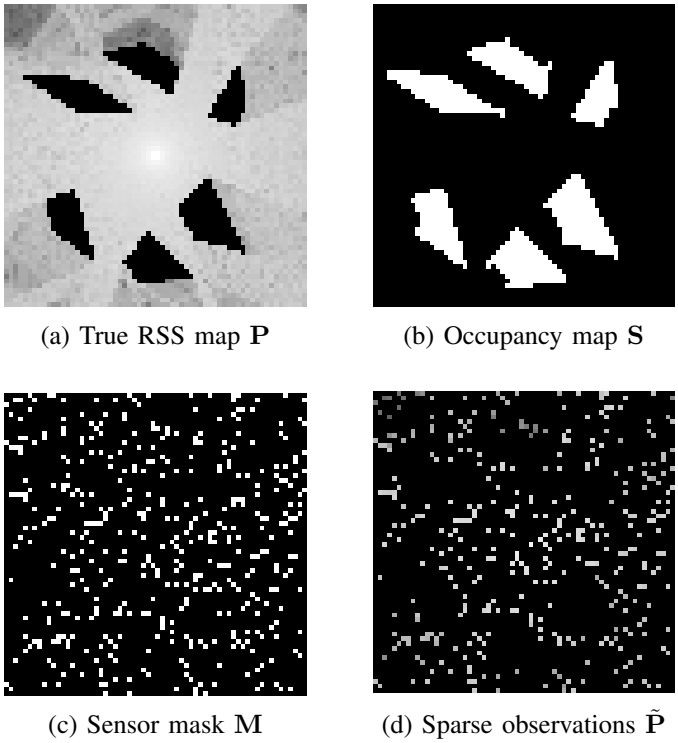
$$\widetilde{\mathbf{P}} = \mathbf{M} \odot \mathbf{P} + \mathbf{\Xi}, \tag{2}$$

where $\odot$ denotes element-wise multiplication and $\mathbf{\Xi}$ is measurement noise, with zero entries outside Ω. The reconstruction model is conditioned on

$$\mathbf{C} = [\widetilde{\mathbf{P}}; \mathbf{M}] \in \mathbb{R}^{2\times L\times W}. \tag{3}$$

Given $\mathbf{C}$, the goal is to estimate the posterior distribution of the dense radio-environment map,

$$p(\mathbf{X} \mid \mathbf{C}) = p(\mathbf{P}, \mathbf{S} \mid \widetilde{\mathbf{P}}, \mathbf{M}). \tag{4}$$



(a) True RSS map $\mathbf{P}$ (b) Occupancy map $\mathbf{S}$

(c) Sensor mask $\mathbf{M}$ (d) Sparse observations $\tilde{\mathbf{P}}$

Fig. 2: Sparse RSS observation model. (a) RSS map $\mathbf{P}$; (b) obstacle occupancy $\mathbf{S}$; (c) sensing mask $\mathbf{M}$; (d) masked RSS $\tilde{\mathbf{P}}$. The task is to recover (a) and (b) from (c) and (d). No measurement noise is added in the experiments.

This reconstruction problem is highly ill posed. At a low sensing ratio $\rho = |\Omega|/(LW)$, observed RSS values are far fewer than unknown RSS and occupancy entries. The unobserved obstacle map must be inferred through

its relationship with sparse RSS patterns. Reconstruction therefore requires measurement consistency, joint modeling of RSS attenuation and obstacle blockage, and efficient inference to support THz ISAC decisions.

IV. Conditional Flow Matching for Radio-Environment Reconstruction

A. Conditional Flow Matching Formulation

We model radio-environment reconstruction as conditional generative transport. Let $\mathbf{X}_1 = [\mathbf{P}; \mathbf{S}]$ be a ground-truth map drawn from the data distribution, and $\mathbf{X}_0 \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ a Gaussian sample of the same dimensions. Flow matching learns a time-dependent velocity field $v_\theta(\mathbf{X}_t, t, \mathbf{C})$ that transports the Gaussian prior to the conditional map distribution, given $\mathbf{C} = [\widetilde{\mathbf{P}}; \mathbf{M}]$. The transport follows the ODE

$$\frac{d\mathbf{X}_t}{dt} = v_\theta(\mathbf{X}_t, t, \mathbf{C}), \quad t \in [0, 1], \tag{5}$$

where $\mathbf{X}_t$ is an intermediate radio-environment state. The learned velocity field permits numerical integration with few function evaluations, motivating its use for efficient THz ISAC reconstruction.

We use a linear interpolation path between $\mathbf{X}_0$ and $\mathbf{X}_1$,

$$\mathbf{X}_t = t\mathbf{X}_1 + (1 - t)\mathbf{X}_0, \tag{6}$$

whose target velocity is

$$\mathbf{U}_t = \mathbf{X}_1 - \mathbf{X}_0. \tag{7}$$

The conditional flow matching objective is then given by

$$\mathcal{L}_{\mathrm{FM}}(\theta) = \mathbb{E}_{t, \mathbf{X}_0, \mathbf{X}_1, \mathbf{C}} \left[\| v_\theta(\mathbf{X}_t, t, \mathbf{C}) - \mathbf{U}_t \|_2^2 \right]. \tag{8}$$

This objective trains the network to predict how an intermediate state evolves toward a radio-environment map conditioned on sparse RSS observations. The condition supplies measurement locations and values, while learned propagation–environment correlations guide reconstruction of unobserved RSS and obstacle occupancy.

To enforce consistency with sparse measurements, we correct the reconstructed RSS channel. Let $\widehat{\mathbf{X}} = [\widehat{\mathbf{P}}; \widehat{\mathbf{S}}]$ denote the generated map. We retain the observed RSS values through

$$\widehat{\mathbf{P}} \leftarrow (1 - \mathbf{M}) \odot \widehat{\mathbf{P}} + \mathbf{M} \odot \widetilde{\mathbf{P}}. \tag{9}$$

The correction leaves unobserved entries unchanged and exactly preserves the available RSS samples. It enforces observation consistency; it does not remove measurement noise. The experiments below use noise-free observations.

B. Joint RSS–Occupancy Reconstruction

The two-channel output jointly performs RSS inpainting and occupancy inference. The mask identifies observed RSS entries, while learned RSS–blockage correlations guide reconstruction of the unobserved occupancy map. Both channels thus contribute to radio-environment awareness.

During inference, we sample $\mathbf{X}_0 \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ and integrate the learned ODE from $t = 0$ to $t = 1$. An Euler solver with N steps uses

$$\mathbf{X}_{t+\Delta t} = \mathbf{X}_t + v_\theta(\mathbf{X}_t, t, \mathbf{C})\Delta t, \quad \Delta t = 1/N. \tag{10}$$

The final generated state gives the reconstructed RSS map and occupancy grid. The experiments use one Euler step, reducing the sampling-step count relative to the diffusion baselines. Its effect on end-to-end ISAC latency remains to be evaluated.

C. Conditional U-Net Velocity Network

We implement $v_\theta(\mathbf{X}_t, t, \mathbf{C})$ with a conditional U-Net. Channel-wise concatenation of $\mathbf{X}_t$ and $\mathbf{C}$ forms a four-channel input. The encoder extracts multiscale propagation and blockage features, and the decoder predicts a velocity field matching the dimensions of $\mathbf{X}_t$. Skip connections retain local information, while time embeddings in residual blocks adapt velocity prediction to different transport stages.

V. Experimental Results

A. Experimental Setup

For training, we sample $t \sim \mathcal{U}[0, 1]$ and $\mathbf{X}_0 \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$, construct $\mathbf{X}_t = t\mathbf{X}_1 + (1 - t)\mathbf{X}_0$, and optimize the U-Net using the flow matching loss in Section IV.

We evaluate the framework on a THz dataset generated with Remcom Wireless InSite ray tracing. Each scenario covers a $100\,\mathrm{m} \times 100\,\mathrm{m}$ outdoor area with a central BS using a 32×32 uniform planar array at 300 GHz and 1 GHz bandwidth. The area contains 1–6 randomly generated quadrilateral obstacles, yielding different blockage and scattering configurations. RSS maps and obstacle occupancy grids are rendered at 64×64 resolution to form two-channel radio-environment maps.

Sparse RSS observations are generated by randomly selecting grid cells as sensing locations. We evaluate $\rho \in \{0.01, 0.05, 0.1, 0.2\}$, spanning ultra-sparse to moderately dense sensing. Selected RSS values are retained, unobserved entries are set to zero, and the binary mask is supplied as an additional conditioning channel. RSS values are normalized to $[0, 1]$ using global statistics. No measurement noise is added, i.e., $\boldsymbol{\Xi} = \mathbf{0}$. Propagation

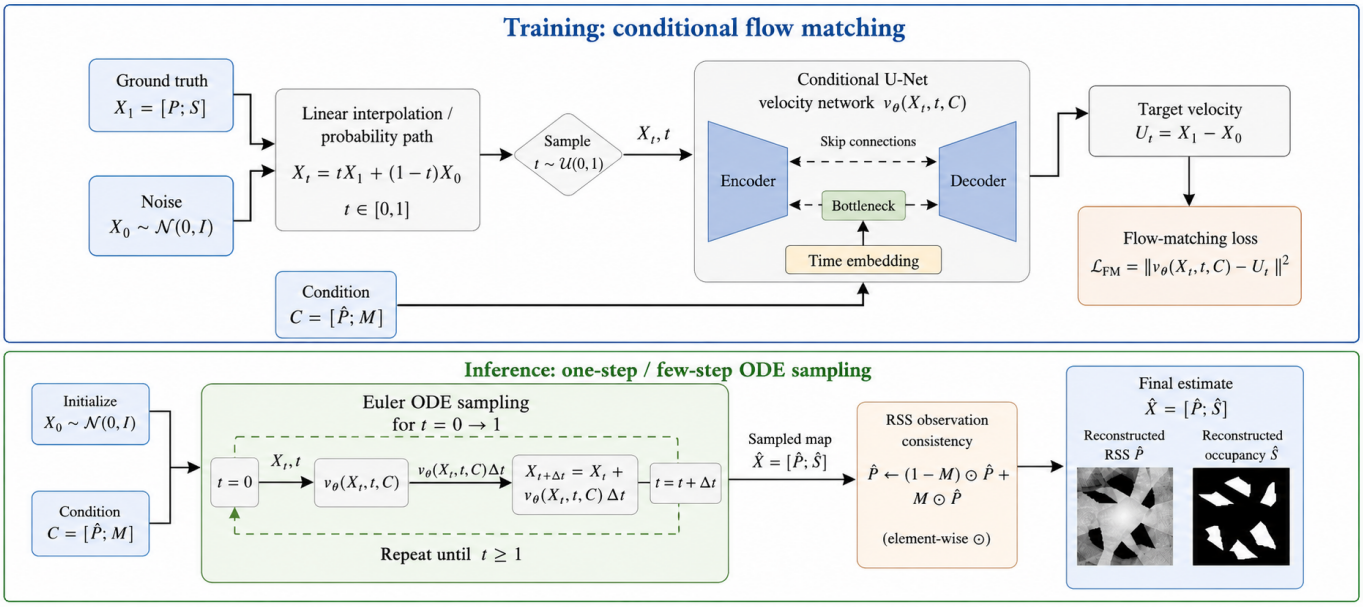


Fig. 3: Conditional flow matching for joint RSS and occupancy reconstruction. Training matches $v_\theta(\mathbf{X}_t, t, \mathbf{C})$ to $\mathbf{U}_t = \mathbf{X}_1 - \mathbf{X}_0$. Inference integrates the learned ODE from Gaussian noise to $\widehat{\mathbf{X}}$, followed by RSS observation-consistency correction.

effects from blockage, reflection, and scattering are represented by the ray-tracing data.

We compare FM with DDPM [8], DDNM [13], and RadioUNet [7]. These baselines represent conditional diffusion generation, diffusion restoration with measurement guidance, and deterministic radio map reconstruction, respectively. All methods use comparable U-Net backbones, the same data split, and the same optimization settings. Sampling steps are selected using validation performance: DDPM uses 1000 steps, DDNM uses 500, and FM uses one unless otherwise specified.

Performance is assessed using mean squared error (MSE), structural similarity (SSIM), and peak signal-to-noise ratio (PSNR), separately for RSS and occupancy. RSS metrics assess radio field reconstruction for coverage estimation and beam management. Occupancy metrics assess obstacle recovery for blockage awareness. These map-level metrics characterize complementary communication and sensing information; downstream control performance is not evaluated here.

B. Performance Comparison under Different Sensing Ratios

Table I compares reconstruction performance across sensing ratios. FM achieves the lowest RSS and occupancy-grid MSE in every tested setting. At $\rho = 0.01$, RSS MSE decreases from 0.1031 for DDPM to 0.0435 for FM, a 57.8% reduction. Relative to the next-best method, RadioUNet, RSS and occupancy-grid MSE decrease by 14.4% and 14.1%, respectively. These results show the benefit of joint conditional reconstruction under sparse sensing.

TABLE I: Performance under different sensing ratios. Bold entries indicate the best value at each ratio; ↓/↑ denote lower/higher is better.

Method	Ratio	MSE↓		SSIM↑		PSNR↑ (dB)	
		RSS	Grid	RSS	Grid	RSS	Grid
DDPM	0.01	0.1031	0.1969	0.2281	0.0131	10.01	7.22
	0.05	0.0565	0.1074	**0.4258**	0.2449	12.90	10.12
	0.10	0.0420	0.0867	**0.5321**	0.1916	15.18	12.19
	0.20	0.0235	0.0576	**0.6195**	0.1747	17.08	13.17
DDNM	0.01	0.1327	0.2123	0.1012	0.0234	9.23	7.13
	0.05	0.0619	0.1070	0.1257	0.0396	12.51	10.24
	0.10	0.0543	0.1081	0.1573	0.0596	13.04	10.31
	0.20	0.0389	0.0667	0.2781	0.1083	14.40	12.03
FM (ours)	0.01	**0.0435**	**0.0741**	**0.3478**	0.0951	**14.18**	**11.95**
	0.05	**0.0259**	**0.0478**	0.3677	**0.2811**	**16.50**	**13.97**
	0.10	**0.0187**	**0.0346**	0.4038	**0.2914**	**17.85**	**15.31**
	0.20	**0.0125**	**0.0274**	0.4768	**0.3274**	**19.53**	**16.34**
RadioUNet	0.01	0.0508	0.0863	0.2334	**0.1108**	13.46	11.30
	0.05	0.0275	0.0499	0.3442	0.2238	16.40	13.78
	0.10	0.0194	0.0361	0.3939	0.1700	17.32	14.57
	0.20	0.0139	0.0278	0.4442	0.3233	18.91	16.13

The SSIM results reveal a tradeoff between metrics. DDPM achieves higher RSS SSIM at $\rho = 0.05$, 0.1, and

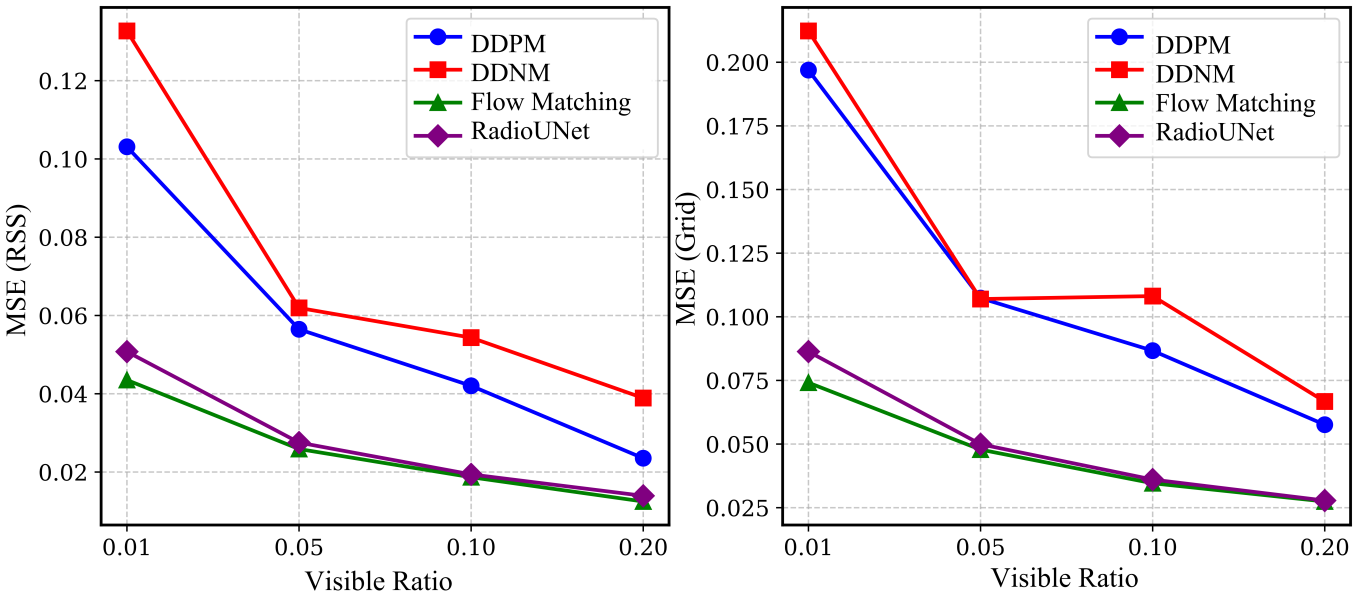
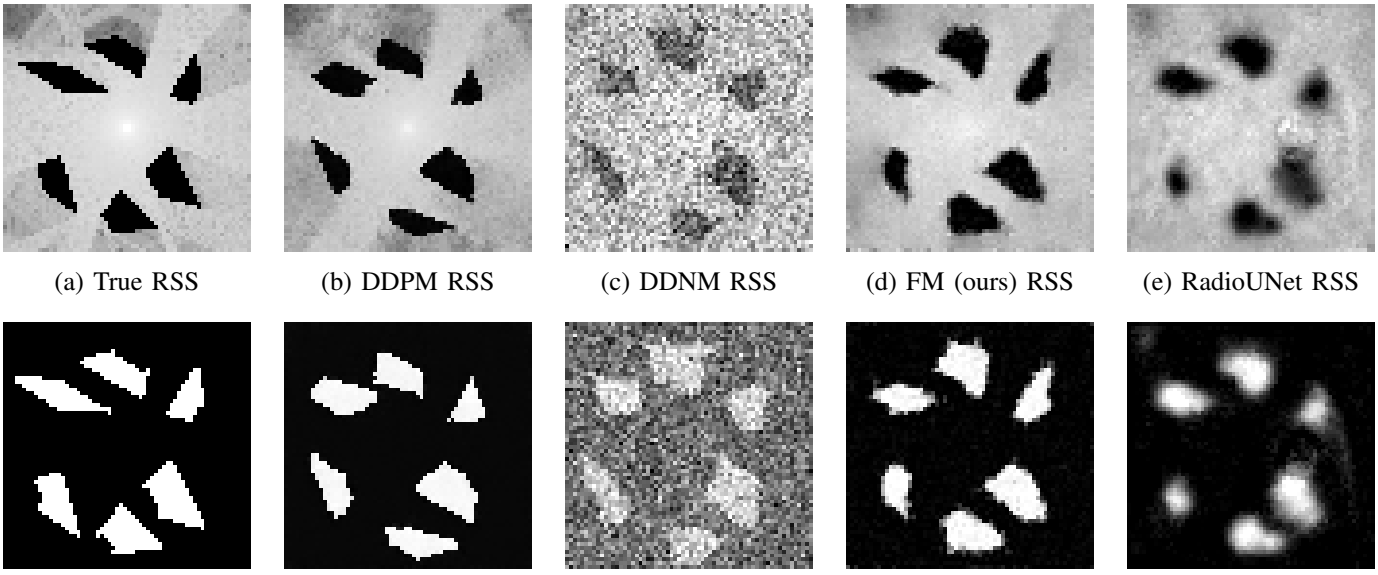


Fig. 4: Reconstruction MSE versus sensing ratio for different methods: left panel RSS MSE, right panel occupancy-grid MSE.



(a) True RSS (b) DDPM RSS (c) DDNM RSS (d) FM (ours) RSS (e) RadioUNet RSS

(f) True grid (g) DDPM Grid (h) DDNM Grid (i) FM (ours) Grid (j) RadioUNet Grid

Fig. 5: Qualitative comparison of reconstructed RSS maps and obstacle grids at sensing ratio $\rho = 0.1$.

0.2, despite its higher MSE and sampling-step count. FM attains the highest occupancy SSIM at these ratios, whereas RadioUNet leads at $\rho = 0.01$. FM also achieves the highest RSS and occupancy PSNR across all tested ratios with one sampling step. Thus, its advantage is consistent for MSE and PSNR, but not universal for SSIM.

Fig. 4 shows the MSE trends with increasing sensing ratio. Errors generally decrease as observations become denser, while FM retains the lowest RSS and occupancy MSE. The largest absolute gaps occur under sparse sensing, where reconstruction is most underdetermined. This regime is relevant to practical THz ISAC because dense sensor deployment can be costly.

C. Qualitative Comparison

Fig. 5 compares RSS and occupancy reconstructions at $\rho = 0.1$. FM reproduces smooth large-scale RSS patterns while retaining local variations around blocked regions. Its occupancy estimate preserves obstacle boundaries with fewer visible artifacts. These examples are consistent with the model exploiting correlations between RSS attenuation and environmental structure. The two channels provide complementary views of the same scene: RSS describes spatial signal variations, while occupancy identifies the obstacles associated with those variations.

These visual differences matter for radio-environment awareness. Blurred RSS peaks can impair link-quality estimation, and distorted obstacle boundaries can affect blockage assessment. DDPM produces smooth RSS maps but less reliable occupancy estimates. RadioUNet tends to smooth object boundaries, while DDNM exhibits occupancy artifacts. FM provides a balanced reconstruction of both channels in the illustrated example, complementing the quantitative map-level comparisons. Preserving both signal variations and obstacle geometry is relevant when these maps serve as inputs to blockage-aware beam selection. The illustrated scene alone, however, does not establish downstream decision accuracy.

Together with one-step ODE sampling, these results indicate a favorable balance between reconstruction accuracy and sampling cost. The tested FM sampler uses one velocity-network evaluation, compared with 1000 DDPM steps and 500 DDNM steps. These step counts describe the sampling procedures rather than measured wall-clock speedups. Direct runtime measurements and downstream evaluations are needed to establish benefits for closed-loop THz ISAC control.

VI. Conclusion

This paper presented a conditional flow matching framework for joint RSS and obstacle-occupancy reconstruction in THz ISAC. Conditioned on sparse RSS observations and sensing masks, the model reconstructs both channels through generative transport. The shared representation combines communication-side radio field estimation with sensing-side obstacle inference, and an observation-consistency correction preserves the available RSS samples. Ray-tracing experiments show lower RSS and occupancy MSE than DDPM, DDNM, and RadioUNet across the tested sensing ratios, using one-step sampling. These findings support FM as an efficient radio-environment reconstruction component. Future work will examine dynamic scenarios and evaluate downstream beam selection and resource allocation.

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